3.16 An SSB-AM signal is generated by modulating an 800 kHz carrier by the signal $m(t) = \cos 2000\pi t + 2\sin 2000\pi t$. The amplitude of the carrier is $A_c = 100$.

1. Determine the signal $\hat{m}(t)$.
2. Determine the (time-domain) expression for the lower sideband of the SSB-AM signal.
3. Determine the magnitude spectrum of the lower-sideband-SSB signal.

3.17 Weaver's SSB modulator is illustrated in Figure P-3.17. By taking the input signal as $m(t) = \cos 2\pi f_m t$ where $f_m < W$, demonstrate that by proper choice of f_1 and f_2, the output is an SSB signal.

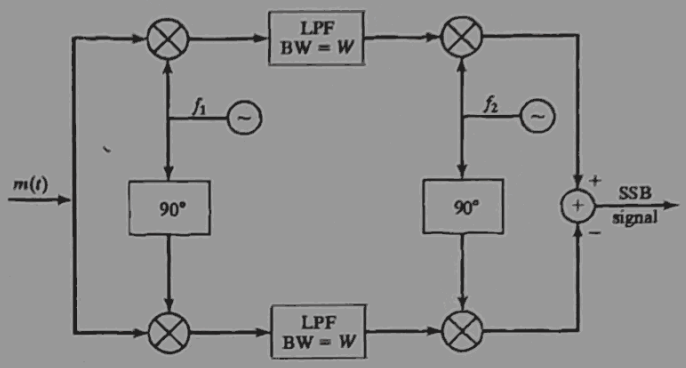


Figure P-3.17

3.24 We wish to transmit 60 voice-band signals by SSB (upper-sideband) modulation and frequency-division multiplexing (FDM). Each of the 60 signals has a spectrum as shown in Figure P-3.24. Note that the voiceband signal is band limited to 3 kHz. If each signal is frequency translated separately, we require a frequency synthesizer that produces 60 carrier frequencies to perform the FDM. On the other hand, if we subdivide the channels into L groups of K subchannels each, such that $LK = 60$, we may reduce the number of frequencies from the synthesizer to $L + K$.

1. Illustrate the spectrum of the SSB signals in a group of K subchannels. Assume that a 1 kHz guard band separates the signals in adjacent frequency subchannels and that the carrier frequencies are $f_{c_1} = 10$ kHz, $f_{c_2} = 14$ kHz, ..., etc.
2. Sketch L and K such that $LK = 60$ and $L + K$ is a minimum.
3. Determine the frequencies of the carriers if the 60 FDM signals occupy the frequency band 300–540 kHz, and each group of K signals occupies the band 10 kHz to $(10 + 4K)$ kHz.

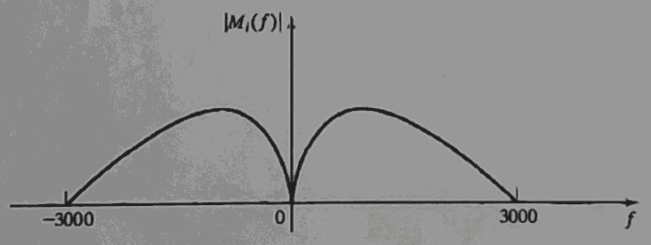


Figure P-3.24

3.16 An SSB-AM signal is generated by modulating an 800 kHz carrier by the signal $m(t) = \cos 2000\pi t + 2\sin 2000\pi t$. The amplitude of the carrier is $A_c = 100$.

1. Determine the signal $\hat{m}(t)$.
2. Determine the (time-domain) expression for the lower sideband of the SSB-AM signal.
3. Determine the magnitude spectrum of the lower-sideband-SSB signal.

1) $\hat{m}(t) = m_n(t) \rightarrow$ hilbert transform

$m(t) = \cos(2000\pi t - \pi/2) + 2\sin(2000\pi t - \pi/2)$

$\hat{m}(t) = \sin(2000\pi t) - 2\cos(2000\pi t)$

2) $y_{SSB} = A_c(m(t)\cos(2\pi f_c t) + m_n(t)\sin(2\pi f_c t))$

$A_c = 100$, $f_c = 800$ kHz, $m(t) = \cos(2000\pi t) + 2\sin(2000\pi t)$

$y_{LSB}(t) = 100(\cos(2\pi 1k\, t) + 2\sin$

sinüsten çıkarılmaya başla

$|a + bj| = \sqrt{a^2 + b^2}$

$y_{LSB}(t) =$

3.17 Weaver's SSB modulator is illustrated in Figure P-3.17. By taking the input signal as $m(t) = \cos 2\pi f_m t$ where $f_m < W$, demonstrate that by proper choice of f_1 and f_2, the output is an SSB signal.

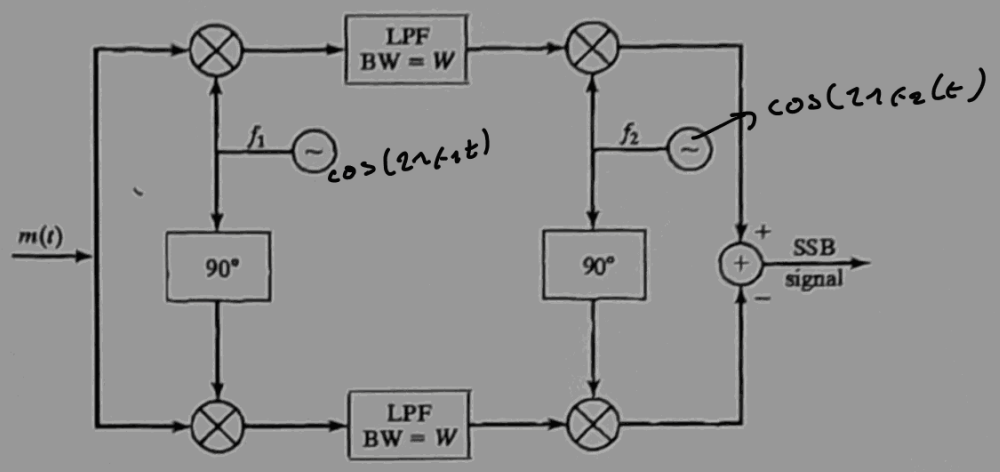


Figure P-3.17

* $|f_1 - f_m| < W$ and $f_1 + f_m > W$

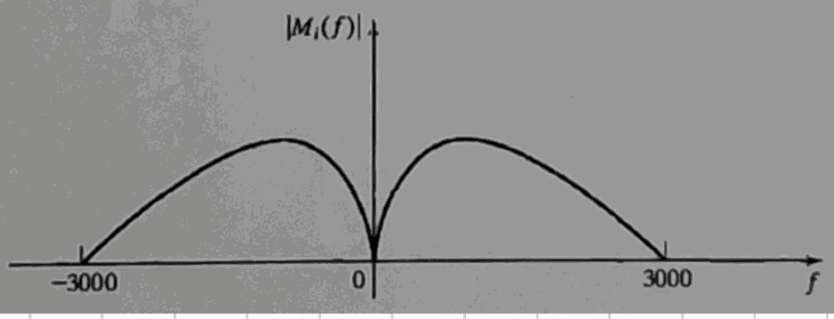
3.24 We wish to transmit 60 voice-band signals by SSB (upper-sideband) modulation and frequency-division multiplexing (FDM). Each of the 60 signals has a spectrum as shown in Figure P-3.24. Note that the voiceband signal is band limited to 3 kHz. If each signal is frequency translated separately, we require a frequency synthesizer that produces 60 carrier frequencies to perform the FDM. On the other hand, if we subdivide the channels into L groups of K subchannels each, such that $LK = 60$, we may reduce the number of frequencies from the synthesizer to $L + K$.

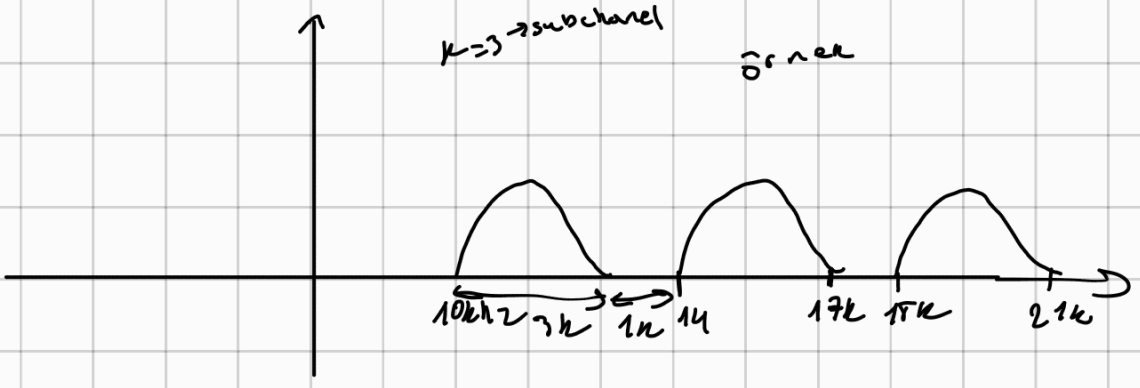
Kalt kanal
L grup

1. Illustrate the spectrum of the SSB signals in a group of K subchannels. Assume that a 1 kHz guard band separates the signals in adjacent frequency subchannels and that the carrier frequencies are $f_{c_1} = 10$ kHz, $f_{c_2} = 14$ kHz, ..., etc.

koruma bandı

2. Sketch L and K such that $LK = 60$ and $L + K$ is a minimum.
3. Determine the frequencies of the carriers if the 60 FDM signals occupy the frequency band 300–540 kHz, and each group of K signals occupies the band 10 kHz to $(10 + 4K)$ kHz.



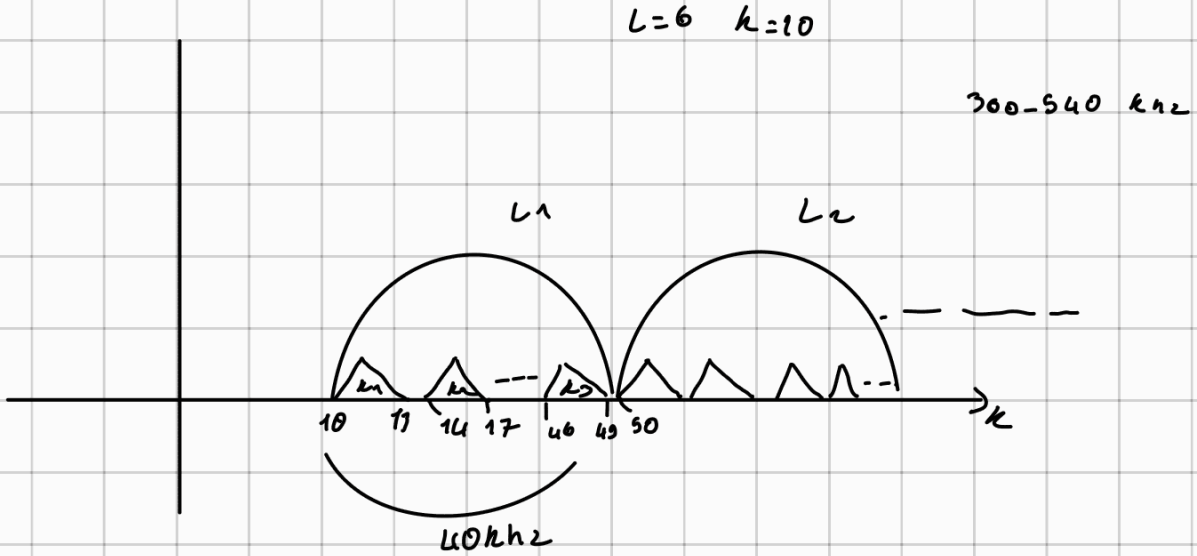


2)

L=6 k=10

300-540 kHz

3)



4.9 The carrier $c(t) = 100\cos 2\pi f_c t$ is frequency modulated by the signal $m(t) = 5\cos 20{,}000\pi t$, where $f_c = 10^6$ Hz. The peak-frequency deviation is 20 kHz.

1. Determine the amplitude and frequency of all signal components that have a power level of at least 10% of the power of the unmodulated carrier component.
2. From Carson's rule, determine the approximate bandwidth of the FM signal.

easy circuits

$P_c = \frac{A_c^2}{2} = \frac{100^2}{2} = 5000$

$P_{I(n)} = \frac{(A_c J_n(\beta))^2}{2}$

4.18 The modulating signal that is the input into an FM modulator is

$$m(t) = 10\cos 16\pi t.$$

The output of the FM modulator is

$$u(t) = 10\cos\left[4000\pi t + 2\pi k_f \int_{-\infty}^{t} m(\tau)\,d\tau\right],$$

where $k_f = 10$. (See Figure P-4.18.) If the output of the FM modulator is passed through an ideal BPF centered at $f_c = 2000$ with a bandwidth of 62 Hz, determine the power of the frequency components at the output of the filter. What percentage of the transmitter power appears at the output of the BPF?

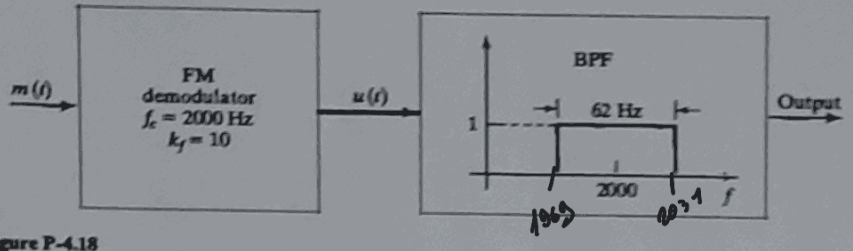
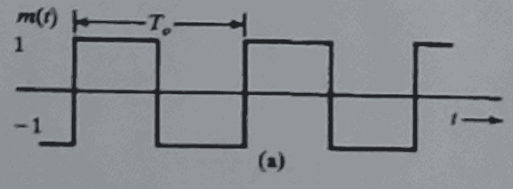


Figure P-4.18

(LATHI)

5.4-2 A periodic square wave $m(t)$ (Fig. P5.4-2a) frequency-modulates a carrier of frequency $f_c = 10$ kHz with $\Delta f = 1$ kHz. The carrier amplitude is A. The resulting FM signal is demodulated, as shown in Fig. P5.4-2b by the method discussed in Sec. 5.4 (Fig. 5.11). Sketch the waveforms at points b, c, d, and e.



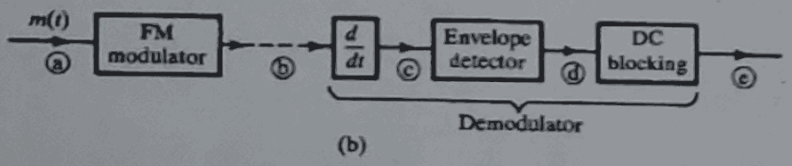


Figure P5.4-2